

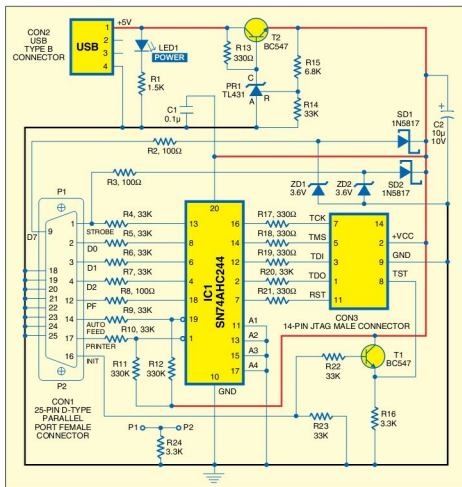


Fig. 6: Circuit of ParaFET Flash emulation for MSP430 MCU

given resistors will do the necessary level translations required between the PC and MSP430 system. JTAG signals TCK, TMS and TDI are driven from the PC through parallel-port pins D0 through D2. While TDO input to the PC is taken through port PF (paper feed on pin 12), STROBE (pin 1) is used to reset the system as well as drive extra power to the target board. Control signals on pins 14, 16 and 17 are used to enable the tristate buffers and also drive a signal called TEST (TST) on the MSP430 system.

Power to the MSP430 target board is supplied through pin 9 (D7) and also boosted using pin 1. These pins, being TTL outputs, can drive a maximum of 2.6 mA at 2.5V. For programming the Flash memory inside the MSP430, typically 1 or 2 mA of current is sufficient. Hence the power from the parallel-port

pins is sufficient to program or erase the MSP430 Flash. The outputs from port pins 9 and 1 are multiplexed using low-power Schottky diodes (SD1 and SD2) after taking across zener diodes ZD1 and ZD2 clamping the voltage to a maximum of 3.6V, for safety reasons.

The 1N5817 Schottky diodes are available in DO41 packages. Colour band on the body denotes cathode. These devices are intended for use in low-voltage, high-frequency inverters, free wheeling, polarity protection and small battery chargers. They exhibit:

1. Very small conduction losses
2. Negligible switching losses
3. Extremely fast switching
4. Low forward voltage drop

One minor drawback of this simple scheme is that the voltage is not well regulated. However, this doesn't affect programming and debugging of most

of the simple circuits.

Due to the current-drive limitation of the parallel port, the users have to ensure that the peripheral logic connected to the MSP430 chip doesn't consume more than 1 mA of current. For cases where the MSP430 system consumes higher current than that supplied by the parallel-port pins, the power supply can be boosted using the PC's USB port. The USB port has 5V supply and can drive up to 100 mA.

A USB cable needs to be connected between the FET and USB port on the PC. USB supply is taken through a precision low-dropout regulator using TL431 voltage reference, which gives a stable voltage of 3V.

The TL431 is a three-terminal precision adjustable shunt regulator, with specified thermal stability over applicable automotive, commercial and military temperature ranges. Its output voltage can be set to any value between Vref (approximately 2.5 V) and 36V, with two external resistors. Typically, the device has output impedance of 0.2 ohm. The active output circuitry

provides a very sharp turn-on characteristic, making the device an excellent replacement for Zener diode in many applications, such as onboard regulation, adjustable power supplies and switching power supplies. Presence of this voltage automatically blocks the supply from parallel-port pins 9 and 1 due to Schottky diodes SD1 and SD2.

Note that the USB connection is required here for driving extra power and regulated supply but not for any signal communication. When the peripheral logic in the target MSP430 board consumes less than 1 mA, the power from the USB port is not required. Hence, power from the USB port is used in the ParaFET when:

1. The target board consumes more than 2 mA and less than 100 mA.
2. The target board requires a well-regulated 3V supply